



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 8	
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Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Dockerfile & Image Build
Implementation & Theory	<pre>from flask import Flask app = Flask(__name__) @app.route('/') def hello_world(): return 'Hello, Docker!' if __name__ == '__main__': app.run(debug=True, host='0.0.0.0')</pre> app.py created — Wrote a minimal Flask app with a single route (/) returning "Hello, Docker!", bound to 0.0.0.0 so it's reachable from outside the container.

```
Flask==2.1.2
Werkzeug==2.1.2
```

requirements.txt created — Pinned dependencies to Flask==2.1.2 and Werkzeug==2.1.2 (pinning Werkzeug explicitly was needed since a newer version broke compatibility with this Flask release).

```
FROM python:3.9-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]
```

Dockerfile created — Used python:3.9-slim as base, set /app as working directory, installed dependencies via pip, copied application code, exposed port 5000, and set the container's default command to run the Flask app.

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 assignment-8 % docker build --no-cache -t my-flask-app .
[+] Building 2.3s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 214B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim-bookworm 0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/6] FROM docker.io/library/python:3.9-slim-bookworm       0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 63B                                    0.0s
=> [2/5] WORKDIR /app                                           0.0s
=> [3/5] COPY requirements.txt .                                0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt     2.1s
=> [5/5] COPY app.py .                                          0.0s
=> => exporting to image                                         0.1s
=> => exporting layers                                           0.1s
=> => writing image sha256:191ebb0cdd154a1d765948619aee827d48f61db2e0e85e227932e7ea13333653 0.0s
=> => naming to docker.io/library/my-flask-app                  0.0s
```

Image built — Ran docker build -t my-flask-app .; all layers built and cached successfully, image tagged as my-flask-app:latest.

```
(base) raghavgdoot@Raghavs-MacBook-Air-2 assignment-8 % docker run --rm -p 5001:5000 my-flask-app
* Serving Flask app 'app' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on all addresses (0.0.0.0)
  WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000 (Press CTRL+C to quit)
192.168.65.1 - - [04/Sep/2026 10:56:37] "GET / HTTP/1.1" 200 -
192.168.65.1 - - [04/Sep/2026 10:56:37] "GET /favicon.ico HTTP/1.1" 404 -
```

Container run + port conflict resolved — Initial run on port 5000 failed (macOS AirPlay Receiver occupies that port by default); remapped to host port 5001 with -p 5001:5000.

Hello, Docker!

Dependency fix + verification — First run hit an ImportError from a Flask/Werkzeug version mismatch; pinned Werkzeug==2.1.2 in requirements.txt, rebuilt the image, and reran successfully — confirmed by accessing <http://localhost:5001> in the browser and seeing "Hello, Docker!"